

Folder 02: Study Selection (PRISMA Process)

1. Overview

This folder contains the documentation for the multi-stage screening and selection process of the SLR titled “*Datasets and Data Preprocessing in Machine Learning-Based Insider Threat Detection: A Systematic Literature Review.*” The selection process followed the PRISMA 2020 guidelines to support transparency, reproducibility, and systematic reporting of inclusion and exclusion decisions.

2. Selection Methodology

The selection process was conducted in four main stages to identify studies with sufficient relevance and technical depth:

- **Stage 1: Identification:** Initial database searches across Web of Science, Scopus, IEEE Xplore, ScienceDirect, and ACM Digital Library identified **639 raw records**.
- **Stage 2: Initial Filtration:** Database filters and manual checks were applied to include English-language, peer-reviewed journal articles published between **2021 and 2025**. This reduced the pool to **397 records**.
- **Stage 3: Screening (Title & Abstract):** The remaining records were managed using Zotero and Rayyan AI. After removing **47** duplicate records, **350** records were screened by title and abstract. At this stage, **226** records were excluded because they were secondary studies, irrelevant to ML-based insider threat detection, or lacked sufficient focus on datasets, preprocessing, or feature engineering. This resulted in **124** articles for full-text retrieval.
- **Stage 4: Eligibility Assessment (Full-Text):** **Seven** articles could not be retrieved in full text, leaving **117** full-text articles for eligibility assessment. These articles were assessed against the inclusion/exclusion criteria and the Quality Assessment (QA) framework. **Seven** studies were excluded after full-text assessment: three did not report datasets or preprocessing details, and four were outside the scope of the review. The final set included **110 primary studies**.

3. Document Descriptions

File Name	Description
README_Study_Selection_and_Filtration.pdf	A folder-level guide explaining the PRISMA-based study selection and filtration process.
PRISMA_2020_Flow_Diagram.pdf	A visual summary of the identification, screening, eligibility, and inclusion stages, including record counts, deduplication, and exclusion numbers.
Selection_Criteria_Notes.pdf	Detailed definitions of the inclusion and exclusion criteria, including how dataset reporting, preprocessing, and general ML relevance were assessed.